

RAHUL NAGARAJAN

AI Engineer

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TECHNICAL EXPERTISE

Agentic Frameworks: LangChain | LangGraph | CrewAI | AutoGen | AWS Bedrock Agents | LangSmith

Agent Patterns: ReAct Agents | Human-in-the-Loop (HITL) | State Management | Tool Calling | Reflection | Multi-Agent Orchestration

RAG & Retrieval: Graph RAG (Neo4j) | Vector DBs (Pinecone, FAISS, ChromaDB) | Agentic RAG | SQL RAG | Hybrid Search | Reranking | Semantic Caching

LLMs & Training: GPT-4 | Claude Sonnet | Llama 3 | Fine-tuning (LoRA, DeepSpeed ZeRO-3) | RLHF | Prompt Engineering

Backend & APIs: FastAPI | Flask | REST APIs

Databases: PostgreSQL | MySQL | Redis | Neo4j

Cloud & MLOps: AWS (Bedrock, Lambda, ECS, SageMaker) | Azure (Bot Service, App Services, Web Jobs) | Docker | Kubernetes | Temporal

Languages: Python | TypeScript | JavaScript | SQL | React

PROFESSIONAL EXPERIENCE

AI Engineer | Muir AI

Seattle, WA • Feb 2026 - Present

- Designed and shipped the pricing pipeline and curated cost dataset that Muir's cost model runs on today; instrumental in landing Muir's first paying customers
- Designed Muir's BoM mass-propagation system to correctly handle customer-supplied weights, quantities, and component counts - upstream of every mass, cost, and emission number Muir ships
- Productionized Muir's LLM-driven price extraction service that grounds specialty materials in real-time market data
- Contributed to Muir's migration to Temporal - decomposed the workflow into typed activities for per-step retries, timeouts, and team-wide visibility; scaled the engine to handle 30+ candidates per product

Software Engineer | Quest Global

Redmond, WA • Sept 2023 - Feb 2026

Aviation Software Analysis (Client: BAE Systems)

- Architected and deployed production agentic system using LangGraph for automated requirements extraction from 500K+ LOC legacy aviation codebases (C/C++), achieving 92% extraction accuracy on validation set of 200 ground-truth test cases labeled by subject matter experts
- Implemented 3-agent workflow (HLR → LLR → TC) with HITL approval gates at each stage, enabling SME review of High-Level Requirements, Low-Level Requirements, and test cases for DO-178C certification workflows across multiple active aircraft modernization programs
- Built state-persistent execution engine with LangGraph StateGraph + PostgreSQL backend, supporting workflow resume across multi-stage processes (averaging 12-15 steps per requirements set) and enabling compliance auditors to replay decision chains for certification documentation
- Reduced requirements engineering cycle time by 65% (from 8 weeks to 2.8 weeks per software module), accelerating time-to-certification and enabling faster deployment of safety-critical avionics updates

Enterprise SQL RAG System with Teams Integration (Internal)

- Architected hybrid SQL RAG system combining Graph RAG (Neo4j for 500+ table schema relationships) with vector search (Pinecone for 200K business rule embeddings), enabling natural language queries with 89% accuracy on complex multi-table joins across enterprise data warehouse

- Designed collaborative agent architecture with specialized roles: Query Router (intent classification), Schema Navigator (graph traversal for table discovery), SQL Generator (text-to-SQL with syntax validation), Result Explainer (natural language response synthesis with source attribution)
- Deployed to Azure production environment (Docker + App Services with auto-scaling 2-20 instances) integrated with Teams via Bot Framework, handling 8K+ queries/week with 1.8s P95 latency and achieving 94% user satisfaction across 10+ internal users in post-deployment evaluation
- Implemented semantic caching layer (Redis) with embedding similarity matching (cosine similarity >0.95 threshold), reducing redundant LLM API calls by 73% and delivering \$4,200/month in cost savings while improving response time by 2.3x for repeated query patterns

Salesforce-Kenandy ERP Integration Platform (Client: 3Sigma)

May 2025 - July 2025

- Architected FastAPI-based ETL pipeline integrating Salesforce CRM with Kenandy ERP system via REST APIs, enabling bi-directional data sync for orders, inventory, and customer records across 15K+ active accounts (processing 350K+ records daily across order, inventory, and customer entities)
- Built Azure Web Jobs for scheduled data synchronization tasks (hourly/daily batches), processing 12K+ daily transactions with automated retry logic, implementing last-write-wins conflict resolution strategy, and data validation pipelines achieving 99.7% success rate
- Implemented Azure SQL-backed queuing system with dead-letter queue handling, idempotency guarantees via unique transaction IDs, and exponential backoff retry logic, maintaining <2 min average sync latency while ensuring zero data duplication across systems
- Delivered \$78K annual cost savings by eliminating 3rd-party middleware subscriptions (\$48K/year for Zapier Enterprise + MuleSoft) and reducing finance team manual reconciliation workload by 60% (from 25 hours/week to 10 hours/week) through automated ETL workflows

EDUCATION

Master of Science in Computer Science | GPA: 4.0/4.0

California State Polytechnic University, Pomona • Aug 2021 - May 2023

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102) - September 2024
- AWS Certified Developer - Associate (DVA-C02) - June 2024
- AWS Certified Cloud Practitioner (CLF-C02) - January 2024